EPSR-test-ex31 mixato: mixing molecule files into a box

Program(s): mixato | chained passes: 1 | verification: ALL IDENTICAL (1 pass)

What this case tests

Exercises `mixato`, the EPSR setup tool [1] that combines single-molecule `.ato` files into one simulation box of given atomic number density, the usual first step in setting up an EPSR simulation of a liquid mixture [2,3]. This basic case types the file names directly, since no directory listing is available.

Code exercised

Fortran (original, fortran_src/):

- `mixato.f`: `mixato` (molecule replication, box size from density, copying of bonds and rotational groups)
- `ato_read_and_save.f`: `minitc` (read `.ato`), `moloutc` (write `.ato`); `ato_manipulation_rev4.f`: `ftype`
- `file_and_directory_commands.f`: `search_for_files` (file prompt / directory listing), `get_full_filename`
- `reallocation_routines.f`: `reallocate1d_r` ... `reallocate4d_i`; `molecom_mod.f90`: `testsize_rotgrps`, `testsize_labrot`

Rust (port, rust/epsr-core/src/):

- `tools/mixato.rs`: `run`, `SavedMol` (first molecule of each file)
- `ato.rs`: `minitc`, `moloutc`, `ftype`
- `tools/setup_common.rs`: `search_for_files`, `get_full_filename`, `command_text`, `read_text`

Physical system

A small dimethyl sulphoxide-water mixture as in the EPSR-test system [4]: 20 DMSO molecules (10 atoms each) and 40 water molecules (3 atoms each), 280 atoms at 0.1 atoms/Å^3 and 300 K, giving a cubic box of side 14.736 Å (`DMSOWaterMix.ato`). As in the Fortran, every copy is placed at the position of the template molecule; randomising the positions is the job of a later program (`randomise`, ex35). Atom types, molecule templates and the random-number state of the last file read are carried over.

Inputs

`DMSO.ato` and `WaterMolecule.ato` copied from EPSR-test; `mixato.stdin`: 2 files (number asked on stdin), `DMSO` with 20 molecules, `WaterMolecule.ato` (name given with its extension) with 40 molecules, density 0.1, output `DMSOWaterMix`. No `system_commands.txt` is present, so the (empty) `system_ls` command fails, `dirlist.txt` is created empty and the names are typed.

How the accuracy was verified

`commands.txt` (1 program call, run once) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; no masking except where noted below). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in 1 pass.

Exercised paths: repeated `minitc` reads accumulating atom and molecule types, copying of bonds and rotational groups, the template-file existence check, the box size from the density and `moloutc` with molecule-scale factors reset to 1 for the added molecules. Both outputs, `DMSOWaterMix.ato` (60 molecules, box 0.14736127E+02 in the reference) and the empty `dirlist.txt`, are identical.

References

[1] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[2] A. K. Soper, "Empirical potential Monte Carlo simulation of fluid structure", Chem. Phys. 202, 295-306 (1996).
[3] A. K. Soper, "Partial structure factors from disordered materials diffraction data: An approach using empirical potential structure refinement", Phys. Rev. B 72, 104204 (2005).
[4] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[5] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.